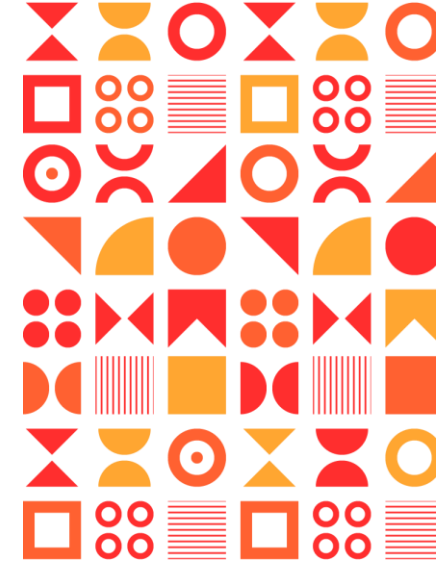




Design Thinking Phase 1: Understand

Semicolon Design Thinking Course

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Learning Objectives

By the end of this module, you will:

- Be familiar with the objectives of the “Understand” design phase
- Be familiar with some methods/tools used in this phase
- Have practiced using some methods/tools related to Understanding
- Have practiced thinking through problem statements and developing design briefs

Understand Phase

In the Understand phase, we work to frame the overall design challenge and research and reflect on overall context for our work.

In this phase, we also review our assumptions about the design challenge topic.

Mindsets for the Understand phase

- Seeing system overviews
- Foreseeing trends
- Reframing problems
- Forming an intent

Methods for the Understand phase

- Defining problem statements
- Reviewing context
 - Desk research
 - Key Facts
 - Expert interviews
 - Convergence maps
 - From-To Matrix
 - Innovation Sourcebook
- Writing a design brief

Framing the design challenge

1. Create an initial definition of the problem you are trying to solve
 - Write it as a short, simple question (for example, “How might we...?”)
 - Summarise intended impact
2. Review context and assumptions
3. Revise the problem statement



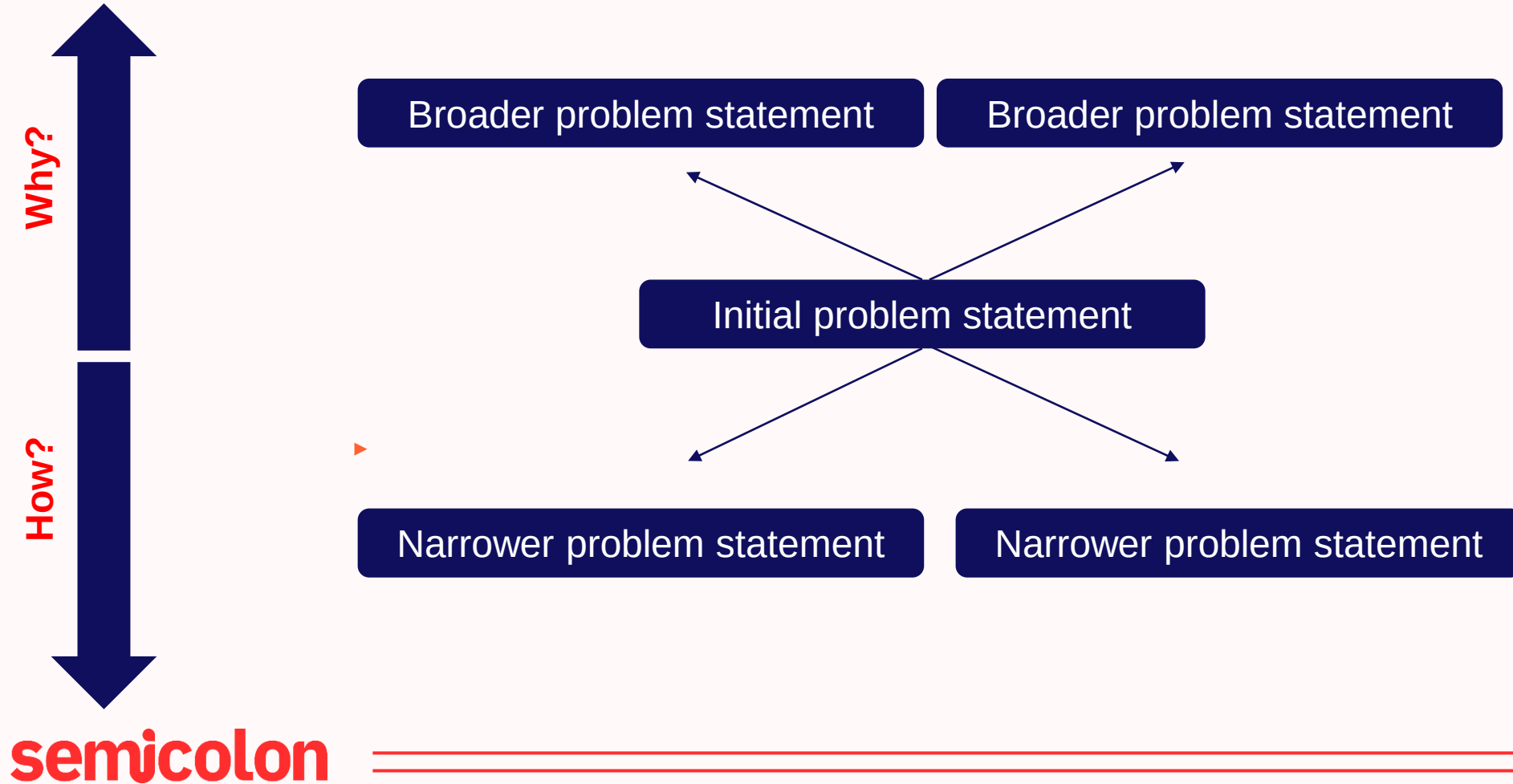
Drafting problem statements

Imagine that you are trying to tackle a challenge related to transportation in Lagos.

Activity:

1. (1 minute) In the middle of a blank piece of paper, write down the design challenge as a single statement, beginning with the words “How might we...”
2. (1 minute) Share the statement with the person next to you.

Exploring levels of problem statements



Methods: Review Context



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Desk research

- Can include activities such as:
 - Internet searches
 - Literature reviews - reading relevant books/journal articles (focusing on evidence)
 - Popular media searches

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Capturing Key Facts

In this method, we outline key facts, or pieces of information relevant to a given topic, in a structured way

Steps:

1. Define the general topic
2. Identify sources of credible information about the topic
3. Conduct research (search widely)
4. Organize information by type
5. Summarize and identify what you have learned from the Key Facts (and what further research may be needed)

source: 101 Design Methods by Vijay Kumar

Activity: Conducting desk research and capturing Key Facts

Activity:

- As a team, work to understand your design challenge topic
- Conduct desk research using a variety of sources
- As you conduct your desk research, create a table capturing Key Facts

Expert interviews

Can deepen our understanding of context for a design challenge and help us understand trends

Process:

1. Identify topics to research
2. Identify experts in that field (eg, professors, authors, industry leaders)
3. Prepare interview questions
4. Conduct the interview
5. Listen and document findings

Convergence maps

- Illustrates how industries or areas of daily life overlap
- Useful when working with trends/changes, to identify new opportunities at the intersection of these topics

Process:

1. Identify topics
2. Identify and summarise trends
3. Build a Venn diagram showing overlapping regions
4. Identify opportunities

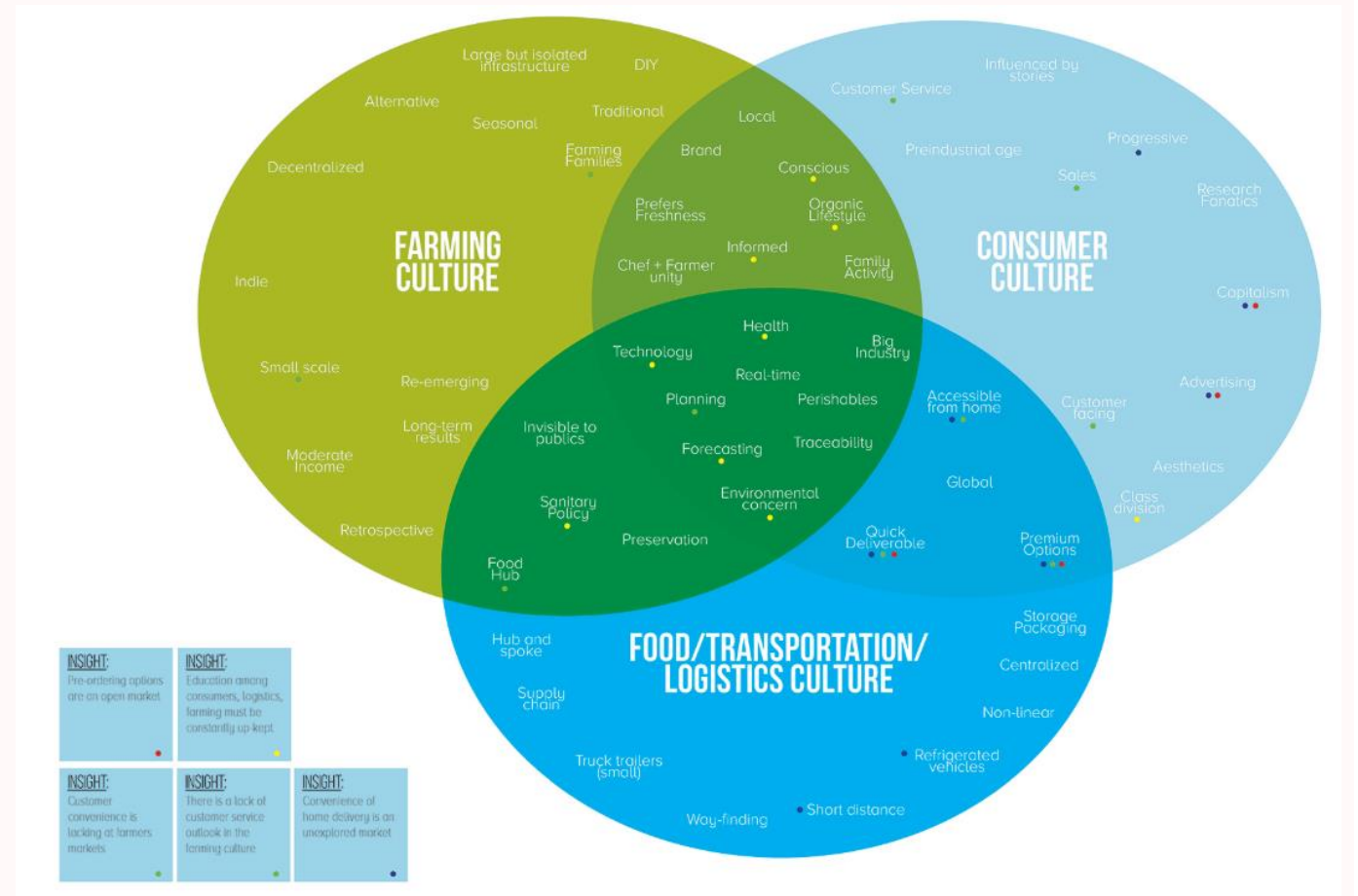


Image source: <http://www.mariadelavega.com/projects/farm-cart-project>

Trends Matrix (From-To Matrix)

- Can deepen our understanding of context for a design challenge and help us understand trends
- By category, map out what happened in the past, what is happening now, and emerging trends/anticipated future scenarios

	Formerly	Currently	Emerging
Users			
Service			
Market			
Culture			
Technology			
....what else?			

Innovation Sourcebook

- In this method, we draw inspiration from existing innovation
- Comparing and contrasting existing examples of innovation helps us understand the reasons they were successful
- Understanding the strategies behind innovation helps us work as innovative teams

Steps:

1. Search for examples of successful innovations
2. Create an Innovation Sourcebook table
3. Compare the innovations to one another. What do the examples have in common? How do they compare in terms of strategic advantage?
4. Think about how the common strategies could be adopted.

Category (offerings, organization, people)	Name	Description	Strategic Advantage

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Preparing design briefs

Documenting the problem statement and some additional plans for the design process cements a common understanding within the team

For design projects undertaken with certain constraints (for example, budgets or deadlines), the brief should also clearly define those parameters

Design Brief – Sample Outline

Why? - Problem

Who? - Target customer

What? - Goals

With what? - Available materials

Who else? - Competition, potential partners

How much? - Budget, other restrictions

When? - Schedules, deadlines

How? - Next steps

Activity: preparing design briefs

Time: 15 minutes

Activity:

Prepare a design brief based on your team's design challenge and selected problem statement. Document the following at a minimum:

- Why? - Problem
- Who? - Target customer
- What? - Goals
- Think about: what already exists? What is missing from existing solutions?

Types of problems

- Well-defined: have a clear solution, but the process of getting to that solution can follow multiple paths.
- Ill-defined: a problem with multiple potential solutions and multiple paths to get to those solutions. Many of the challenges faced in our day to day work would fall under this category. For example, “How might we make our app more appealing to users?”
- “Wicked problems”: are complex and ambitious. A social or cultural issue or concern that is difficult to even explain, let alone solve. Usually, the problem itself is unclear, so we may go through several stages just to get to a problem statement defined.

These are the biggest design challenges - and can lead to the greatest impact!

Key Learning Points

- In the Understand phase, we think broadly about a design challenge, working to understand context and trends and identify opportunities for change
- In addition to conducting general desk research, using some structured methods such as a Trends Matrix can help teams think through context of a design challenge
- Starting with a clear problem statement (for example, in “How might we...” format) and a design brief can give design teams a good sense of direction
- Defining the problem may be challenging, particularly when tackling ambitious/complex problems. If teams are stuck debating the problem statement, have a “bias for action” and get started - the problem statement can be refined as work gets underway



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